Deployment Guide & Bill of Materials (DEPLOYMENT_AND_BOM.md)

1. Observed Hardware & Environment Inventory (hardware_inventory.json)

Component	Observed Identity & Specifications	Connection / Role
Primary Development & Demo Host	Apple MacBook Pro (MacBookPro18,1, Apple M1 Pro arm64, 10 cores, 16 GiB RAM, macOS 27.0.1, Python 3.13.15)	Runs full test suite, 1 Hz simulator, fail-closed evidence verifier, PDF generator, single-screen 2D Operator UI (http://127.0.0.1:8765), and synchronized Three.js 3D Vessel Twin (viz3d/).
Remote Edge Linux SBC (ssh ardu / ventuno)	Arduino Ventuno Q (training06, arduino,monza / Qualcomm QCS8300 Monza 8-core aarch64: 4x Cortex-A78C + 4x Cortex-A55, 14 GiB RAM, 55 GB eMMC, Ubuntu 24.04.4 LTS noble, Kernel 6.8.0-1080-qcom, Python 3.12.3)	Connected via Wi-Fi LAN (10.0.0.130). Hosts isolated /home/arduino/sdv-resource-governor/.venv (--system-site-packages) for edge deterministic replay (G1+G3+G4+G5 Scenario B, 9,560 kg, 9,853 kg-eq) and verified semantic parity (PASS).
USB Serial Interface	/dev/cu.usbmodem1C49DF7200002 (MEGAHUNT VCPHID, VID 0x0483, PID 0x5750)	Enumerated over USB-C. Note: Ventuno Q ADB USB enumeration requires external 7-24 V DC barrel/header power; when unpowered, sdv-governor doctor reports adb_usb: BLOCKED_CONNECTION without affecting standalone or LAN operation.
STM32H5 Real-Time MCU	Onboard MCU co-processor (arduino:zephyr:ventunoq)	Out of scope for the admission/arbitration challenge; no UNO Q motor sketch is flashed.

2. Software Bill of Materials (SBOM -- Synchronized with THIRD_PARTY_COMPONENTS.csv)

Package / Tool	MacBook Pro (requirements-macos-arm64.loc	Arduino Ventuno Q (requirements-linux-aarch64.lock)	License	Purpose
Python	3.13.15	3.12.3	PSF-2.0	Standard library (fractions, dataclasses, hashlib, http.server, xml.etree.ElementTree)
PyYAML	6.0.3	6.0.3	MIT	Declarative YAML loading for ship baseline, guest manifests G1-G6, priority profiles, scenarios, and errata/clarification overrides
pytest	9.1.1	9.1.1	MIT	Automated test execution and JUnit XML generation (junit.xml)
jsonschema	4.26.0	4.10.3	MIT	Formal JSON Schema Draft 2020-12 validation of all 7 schemas (schemas/*.schema.json)
attrs / referencing / rpds-py / pyrsistent	26.1.0 / 0.37.0 / 0.30.0	23.2.0 / 0.20.0	MIT	Transitive dependencies of jsonschema
three.js	0.160.0 (viz3d)	N/A (browser UI)	MIT	WebGL 3D rendering library for the synchronized 3D Vessel Twin (viz3d/) bound to /api/state
vite	5.x (dev-only)	N/A	MIT	Offline frontend bundler used only during development to build viz3d/dist
Blender	4.x (offline tool)	N/A	GPL-3.0	External offline 3D asset authoring tool for static GLB meshes; not linked or invoked at runtime

(Zero ML/LLM/vision dependencies (onnx, torch, opencv, transformers, gguf) are installed or imported in .venv.)

3. Clean / Offline Setup & Remote Board Commands

```
# Clean environment setup with dev + schema validation dependencies
python3 -m venv .venv --system-site-packages
.venv/bin/pip install -r requirements-macos-arm64.lock
.venv/bin/pip install --no-build-isolation --no-deps -e .

# Run local hardware & transport doctor
PYTHONPATH=src .venv/bin/python -m sdv_governor.cli doctor

# Sync codebase to remote Linux board (when reachable on ssh ardu) and verify semantic hash parity
PYTHONPATH=src .venv/bin/python -m sdv_governor.cli deploy-board --ssh-target ardu
```